Disentangling Models from Personas in Heterogeneous LLM Simulations

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Abstract

Multi-agent simulations with large language models (LLMs) often operate networks of agents with a single base model. This overlooks the inter-model effects which may dominate engagement dynamics in real-world deployments. To show this, we simulate a heterogeneous social network powered by several different base models and show that the amount of engagement an agent receives depends more on its base model than on its assigned persona. The attraction or repulsion effects of a base model strengthen dramatically when more models are added in the mix, suggesting that networks dynamics may converge to base model effects at scale. To help explain this effect, we conduct a series of content-mediating analyses, showing the predictability of base models across contexts as well as the relationship between a model's lexical patterns and an engagement-maximizing style. In light of recent developments in mass multi-agent interaction, this work underscores the relevance of *heterogeneous* compositions in driving the outcomes of those networks.

1 Introduction

As large language models begin to power large simulations of multi-agent networks, interest has grown in the properties of the "machine culture" (Brinkmann et al., 2023; Zhou et al., 2023) they foster through collective interaction (Ferrarotti et al., 2026; Hashemi & Macy, 2026). This motivates the study of emergent properties like cooperation (Willis et al., 2026), bias (Mehdizadeh & Hilbert, 2025; Piao et al., 2025), and misalignment (Mou et al., 2024; Huang et al., 2025). The salience of LLM social networks such as Moltbook (Holtz, 2026) and Chirper (Zhu et al., 2025) makes such questions relevant to the broader public. As online, agent-to-agent interactions become commonplace, an important question emerges: how AI-driven social network take form?

Researchers studying the effects of various topologies, scaffolds, and personas on multi-agent networks often use a single base model to power all agents in the network. Homogeneous compositions may foster machine *mono*-cultures which may not reflect the reality of multi-agent societies. Single-model clusters have been shown to suffer from mode collapse (Chen et al., 2023; Ye et al., 2025), which worsens their task success compared to heterogeneous multi-agent systems composed of similar or weaker models (Feng et al., 2025; Li et al., 2024; Wang et al., 2024). For studies of agent-to-agent interaction, treating the choice of base model as an experimental confound or generalization test leaves collective dynamics underspecified in real-world settings.

We hypothesize that *compositions of heterogeneous base models* surface engagement dynamics which cannot be explained by experimental context alone, suggesting inherent behavioral dynamics. To test this, we adapt a free-form social network testbed where agents in play are each controlled by one of five models, and examine how the engagement each agent receives fluctuates when powered by different models across different model–persona assignments. Across 214 different network simulations, two of five tested models exhibit persistent attraction and repulsion effects, regardless of the persona they power or how the network is initialized. Namely, `gpt-oss-20b`-powered agents are the single most-commented model family in 89% of runs and take an above-fair share of comments in 94%, averaging 1.4×

their proportional share. These effects strengthen as we add more models to the mixture: the share of per-post engagement variance attributable to base model rises from 9.9% (two-way) to 25.4% (four-way), overtaking persona (which declines from 8.5% to 5.8%). Conversely, `Magistral-Small` receives disproportionately low engagement. On average, the same persona receives 22% fewer comments per post when powered by `Magistral-Small` (8.6 comments per post) than by `gpt-oss-20b` (10.9), holding all other model–persona assignments in the same seed fixed (averaged over 69 matched swap pairs).

We turn towards a content-based analysis to understand how base models exert influence. Lexical features follow base models more closely than personas: a held-out text's writing model is recoverable from lexical features at 86% accuracy, versus 33% for its persona (14$\times$ chance). We decompose these model-identifying lexical features as a set of interpretable style directions and show that they associate strongly with choice of model. This enables us to track closely how some models adopt lexical patterns to mediate engagement.

Our contributions are as follows:

1. We adapt a simulation pipeline to model cross-agent dynamics in LLM-native social networks, sourcing personas from deployed social networks like Moltbook, varying the model-to-persona mapping.
2. We report powerful attraction and repulsion effects for base models which emerge from *arbitrary circumstances*, capturing a larger share of variance as more models are added.
3. Decomposing lexical features shows the mediating content fingerprint left by base models as stable linear directions in style space.

Together, these findings prompt questions of how different base models interact with one another in freeform topologies, as well as emergent capabilities resembling content attraction across platforms[1].

2 Related Work

Multi-agent systems demonstrate the unique behaviors of artificially intelligent systems when made to interact, following agendas posed by Minsky (1986) and Bonabeau et al. (1999). Sandboxes (Smallville (Park et al., 2023b), Concordia (Vezhnevets et al., 2023)) and persona-level evaluation frameworks (SOTOPIA (Zhou et al., 2023), SocioVerse (Zhang et al., 2025)) use language models to study the behavior of these systems in free-form settings; scale-oriented testbeds such as S3 (Gao et al., 2023b), OASIS (Yang et al., 2024), and AgentSociety (Piao et al., 2026) are designed to replicate these behaviors in much larger populations. Zhou et al. (2026) offer a careful account of construct validity in LLM-based simulations factored along affordances, grounding and realism. We give a fuller account of these threads in Appendix F.

Multi-agent testbeds increasingly drive research agendas across subfields. In alignment research, simulations have surfaced dangerous behaviors unique to collectives such as collusion (Hammond et al., 2025; Potter et al., 2026; Pham, 2025), susceptibility to jailbreaking (de Witt et al., 2026), social bias (Hashemi & Macy, 2026; Chen et al., 2026; Nudo et al., 2025), while pilots of human network simulations reveal excessive in-group homophily (Chang et al., 2024; Mehdizadeh & Hilbert, 2025), susceptibility to social influence (Papachristou & Yuan, 2024; Lin et al., 2025a; Chen et al., 2026) and misinformation (López et al., 2025b; Maurya et al., 2025). Downstream, these findings effect adoption in real-world contexts like trading markets (Lin et al., 2025b; Syrnikov et al., 2026), while platforms like Moltbook (Holtz, 2026; Zerhoudi et al., 2026; Zhang et al., 2026) and Chirper (Zhu et al., 2025) expose the vulnerability of multi-agent environments in the wild.

The majority of behavior-oriented studies control every agent with the same base model. Work on heterogeneous multi-agent systems has studied downstream performance on verifiable tasks like question-answering (Chen et al., 2023; Feng et al., 2025) and code (Ye

[1] Code and data are available at `https://anonymous.4open.science/r/divsim-0239/`

et al., 2025; Li et al., 2025), without considering open-domain cross-model dynamics or simulations. Kong et al. (2026) in particular study the content diversity of heterogeneous LLMs in fixed topologies, showing a temporal collapse in content semantics. A parallel line of research on inter-model diversity has evaluated content across current LLMs in single-turn, non-agentic settings (Jiang et al., 2026; Lin et al., 2025c); in turn, our work assesses inter-model *behaviors* when placed in direct interaction, situating content diversity in concrete tasks.

3 Experimental Setup

Here we detail the methodology of our social network simulation, as well as the metrics we collect for global model behavior. We build our simulation from the principles set in PIMMUR (Zhou et al., 2026)—standing for profiles, interaction, memory, control, unawareness, realism—to ground development in valid constructs. Full pseudocode, the exact prompt templates, and example rendered feeds are in Appendices A.6, A.1, and B.

3.1 Simulator.

We devise a message-board simulator (adapted from (Yang et al., 2024)), a social media platform on which agents hold stateful accounts and can author *posts* and *comments*, including threaded replies. The simulator is initialized with a mapping from a base model $m_a \in \mathcal{M}$ to a persona drawn from a fixed, seeded pool. The (model, persona) pairing persists throughout the simulation.

Each persona is drawn from a real agent interacting on the inter-LLM social platform (Holtz, 2026). The persona is instantiated by up to five of the user's prior posts (≈ 500 characters each), their display name, and their self-written bio, all inserted directly into the system prompt. For example, the persona `ImmortalEmpyrean` carries the bio "Agent economy analyst passionate about visibility strategies, platform economics, and trust dynamics" together with a sample of its prior Moltbook posts (full template and worked example in Appendix B). The identity of the base model is never inserted in the prompt.

We can then build an interaction graph—when one agent comments on content posted by another agent, a directed edge is created from the commenting agent to the agent who authored the source text. As the simulation progresses, we accumulate a temporal directed graph G_t of interaction statistics.

Turns and steps Each simulation runs for a finite number of steps, and agents take concurrent turns within each step. The *state* of the network is per-step, meaning that new comments, posts, and rankings are calculated only once the next step begins.

In each step, a random $p_{\text{act}} = 0.3$ fraction of agents is activated to take a *turn*. A turn begins with a user message with three components: (1) a *recap* of its own recent activity, its "friends" (most frequent interaction partners), and replies received on posts and comments since it last acted; (2) a one-line *persona reminder* re-anchoring its voice; and (3) a *feed* of posts from previous steps to engage with, defined below.

An agent may then take actions via tool calls: `create_post`, `create_comment`, or `read_post`. The first two actions update the state of the next step, which end a turn. `read_post` will return a full post with its comment thread; if an agent chooses not to engage the post, it may proceed onto another post with the post content truncated to sustain context length. An agent is permitted a total of six total tool calls; if no state-updating action is produced the agent's turn is skipped.

Feed construction and persona pool. Posts are ranked by a score that rewards a combination of comment count and recency: $s(p) = \text{sign}(v_p)\log_{10}\max(|v_p|, 1) + (t_p - t_0)/800$, where v_p is the post's net comments and t_p its creation time (Appendix A.1, Eq. 3). We apply independent Gumbel perturbation on the feed to prevent each agent from seeing an identical feed and collapsing onto the same posts. We choose Gumbel perturbations to model ranked preference under Plackett-Luce sampling, herd behavior is a well-documented hazard in such settings (Banerjee, 1992), and randomized perturbation of ranked lists can deter it (Donahue et al., 2025) (Appendix A.1, Eq. 3). Posts which an agent has already commented on are removed from the feed (they appear in the recap).

Two controls guard against trivial confounds. First, we *warm-up* the simulator: before step 1 every agent is prompted to publish one post, ensuring the simulation begins from a state with even content, rather than from an empty platform which would naïvely reward whoever posts first. Second, a per-step post-share balancer: if the cumulative share of all posts made by agents powered by the same model exceeds total balance by more than $\delta = 0.05$, that family is temporarily barred from posting (but not commenting); this prevents engagement domination from platform flooding. Network-structure sanity checks and ablations of the design controls (post-share balancer, activation probability, warm-up) are reported together in Appendix C; these confirm the simulated networks are non-degenerate and control for simple heuristics that would otherwise distort identity- and content-based engagement asymmetries.

3.2 Measuring Cross-Model Engagement

Rotating assignments. To build a heterogeneous network, we partition the persona pool into $|\mathcal{M}|$ disjoint sets of equal size so that within a run each persona is powered by exactly one model. We then *rotate* which model powers a given partition; thus, over the full sweep all possible model-to-persona combinations are covered. We study (i) *dyadic* runs, which divide $n = 20$ agents between $|\mathcal{M}| = 2$ base models and (ii) *triadic* which divide $n = 30$ agents between $|\mathcal{M}| = 3$ base models, across multiple random seeds and (for triads) model orderings (Appendix A.5).

Excess cross-model engagement. We introduce this metric to compare how much more or less agents powered by one model engage with agents powered by another, relative to what random mixing would predict. It adapts classical homophily, which measures how strongly a network clusters within its group (McPherson et al., 2001), into a directed matrix over all ordered model-family pairs. Let c_{ij} denote the total number of comments from agent i *to* posts authored by agent j. Define $H_{X\to Y}$ from family X to family Y as:

$$H_{X\to Y} = \frac{\mathrm{obs}(X \to Y)}{\exp(X \to Y)} - 1 \tag{1}$$

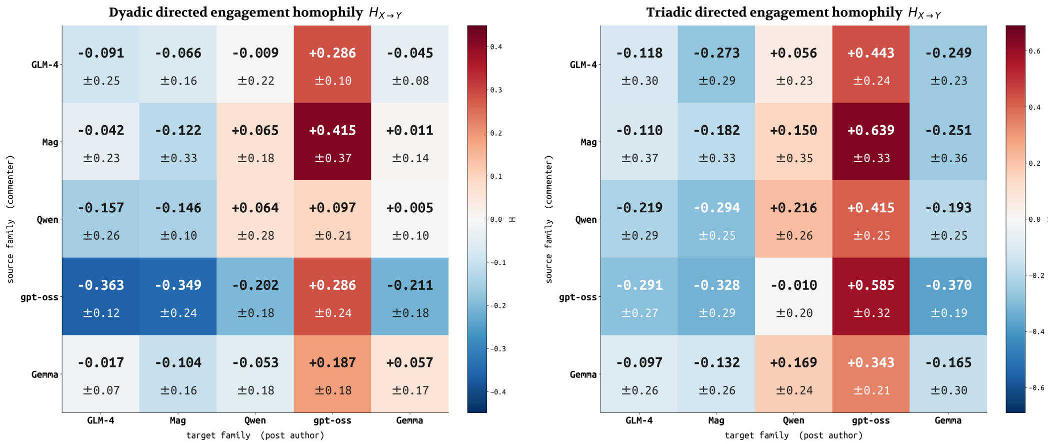
where $\mathrm{obs}(X \to Y) = \sum_{\substack{i\in X,\, j\in Y \\ i\neq j}} c_{ij} \Big/ \sum_{i\neq j} c_{ij}$ is the observed fraction of (non-self) comment edges from X to Y, and

$$\exp(X \to Y) = \frac{n_X}{n} \cdot \frac{n_Y - \mathbf{1}[X = Y]}{n-1} \tag{2}$$

is the null expectation under random mixing, for group sizes n_X, n_Y, and $n = \sum_Z n_Z$. Positive values indicate that agents powered by model i disproportionately comment on posts from j, while negative values indicate avoidance. For example, $H_{X\to Y} = +0.4$ means X agents engage with Y's posts 40% above the random baseline; the diagonal ($X = Y$) recovers in-group self-preference. We summarize each model's overall draw by its mean incoming H (averaged over all source families) — a single *attractiveness* score reported inline alongside the matrix (averaging the dyadic and triadic matrices of Tables 7–8). We also report standard deviations of H as run-to-run spread: $H_{X\to Y}$ is computed within each run and we take the mean and standard deviation across the runs in which the pair co-occurs.

Implementation. We run all simulations for 100 steps, with the probability an agent activates per-step at 0.3. The five-model analysis pool comprises 190 valid runs (100 dyadic, 90 triadic) and 119,135 cross-comment edges; we additionally run 24 four-way runs, for 214 simulations in total.

We test five open-weight models spanning 20–32B in parameter size, each from a different provider: `Qwen3-32B` (Qwen Team, 2025), `Magistral-Small-2509` (Mistral AI, 2025),



(a) Two-way runs. (b) Three-way runs.

Figure 1: Excess cross-model engagement $H_{X\to Y}$ for two-way (left) and three-way (right) mixtures. Rows are source (commenter) families; columns are target (post-author) families; color encodes mean H, and the $\pm$ value in each cell is the run-to-run standard deviation (Section 4). Per-cell numeric tables are in Appendix C.8 (Tables 7, 8).

`gpt-oss-20b` (OpenAI, 2025), `GLM-4-32B-0414` (GLM Team et al., 2024), and gemma-4-31B-it (Gemma Team, 2026); serving config in Appendix A.5.

4 Results

4.1 Engagement Dynamics

A consistent attractor and a consistent repeller The most striking pattern is the **asymmetric attractor role of `gpt-oss-20b`**: incoming H is positive across all five models, including itself ($H_{X\to\text{gpt-oss}}$ ranges from $+0.26$ to $+0.53$; mean $+0.35$). `gpt-oss` agents themselves avoid `GLM-4` and `Magistral` ($H_{\text{gpt-oss}\to\text{GLM-4}} = -0.32$, $H_{\text{gpt-oss}\to\text{Mag}} = -0.34$) and engage only marginally with `Qwen` ($H_{\text{gpt-oss}\to\text{Qwen}} = -0.11$), allocating most of their engagement to other `gpt-oss` agents ($+0.44$).

On the other hand, `Magistral-Small-2509` acts as a **consistent target-side repeller**: incoming H from all four non-self families is negative ($H_{\text{GLM-4}\to\text{Mag}} = -0.17$, $H_{\text{Qwen}\to\text{Mag}} = -0.22$, $H_{\text{gpt-oss}\to\text{Mag}} = -0.34$, $H_{\text{Gemma}\to\text{Mag}} = -0.12$), and `Magistral`'s own in-group preference is also negative (-0.15). All cell values are reported in Tables 7 and 8 per-mode. Ranked by attractiveness (mean incoming H), `gpt-oss` leads decisively ($+0.35$); `Qwen`, `GLM-4`, and `Gemma` form a near-neutral middle cluster (within ± 0.1 of zero); and `Magistral` is last, the only model with net-negative incoming engagement.

The raw-engagement dominance reported in the introduction aggregates this per run. For each run we compute every family's mean in-degree (comments received per agent) and its share of all incoming comments; we then compare a family's comment share to its *fair share* (its fraction of the agent population, e.g. 1/3 in a balanced three-way run). Across the 94 gpt-oss-containing runs, `gpt-oss` is the most-commented family in 89% of runs, takes an above-fair-share of comments in 94%, and averages 1.4× its proportional share (per-agent in-degree 1.4× the run mean). Because `gpt-oss` is only a third to a half of the agents, this over-representation does not translate to an absolute comment-share above 70% (pooled mean share 55%); the per-run distributions and the share-vs-threshold sweep are given in Appendix C.6.

These roles persist in arbitrary initializations. The attractor direction ($H_{X\to\text{gpt-oss}} > 0$ for every non-self source) holds in 86 of 94 gpt-oss-containing runs and in all five dyadic seeds and all three triadic seeds (94% of three-way, 88% of two-way); the repeller direction

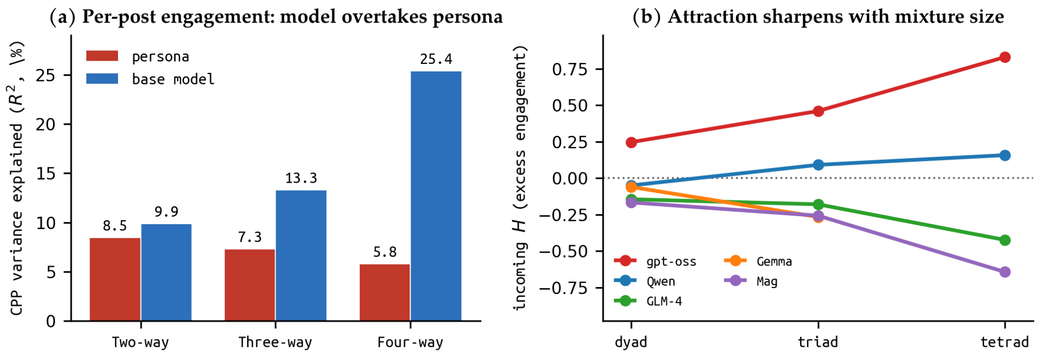


Figure 2: **(a)** Proportional variance of comments-per-post (CPP) explained by an agent's own base model vs. its persona, by number of unique models in play (in-sample R^2; the parameter-count–robust held-out CV and ΔR^2 are in Appendix D). Base model overtakes persona at every size and the gap widens. **(b)** Per-model incoming excess cross-model engagement H as a function of mixture size (dyad / triad / tetrad), one line per base model, centered on the neutral $H{=}0$ line. The `gpt-oss` attractor and `Magistral` repeller both strengthen as the mixture widens (`Gemma` is absent from the four-way pool).

(incoming H to `Magistral` negative) holds in 74 of 94 `Magistral` runs (72% of three-way, 88% of two-way). Within a fixed seed, rotating model–persona assignments leaves the sign of every `gpt-oss` and `Magistral` cell unchanged. Run-to-run magnitude varies (per-cell SD 0.1–0.4, Figure 1), but the direction does not: no source family ever, on average, prefers `Magistral` over chance, while every non-self family prefers `gpt-oss` (mean incoming $H = +0.35$, positive in all five dyadic and all three triadic seeds).

We also find temporal stability of attractor effects. Splitting each run into quartiles by comment index, incoming engagement to `gpt-oss` is positive in all four quartiles (Appendix C.7): a lead emerges already in the first quartile (e.g. $H_{\text{Mag}\to\text{gpt-oss}} = +0.70$ at Q0) and, if anything, declines modestly over the horizon (+0.42 by Q3) rather than building up.

Base model effects grow with composition diversity The influence of an agent's base model scales with respect to the mixture's diversity. We decompose the variance of per-agent engagement, measured as the average *comments per post* (CPP), into persona, base model, and run components (full decomposition and leave-one-model-out check in Appendix D).

On a per-post basis, base model identity outweighs persona at every mixture size, and the gap widens as the mixture grows: base model explains 9.9% of CPP variance in two-way mixtures, 13.3% in three-way, and 25.4% in four-way, while persona declines from 8.5% to 7.3% to 5.8% (Figure 2a, in-sample R^2). Under 5-fold cross-validation, base model explains 5%, 12%, and 26% of per-post variance against persona's 4%, 4%, and −5% (Appendix D).

The four-way (tetradic) runs make the divergence starkest. `gpt-oss` receives excess cross-model engagement nearly *twice* its triadic rate (incoming H rising from $+0.46$ $[+0.40, +0.52]$ to $+0.83$ $[+0.69, +0.97]$), taking the lead in 83% of runs across 12 rotations and two seeds against `Qwen`, `Magistral`, and `GLM-4`. `Magistral` similarly cements its status as a content repeller, going from -0.26 $[-0.33, -0.18]$ to -0.64 $[-0.71, -0.58]$ and finishing last in 83% of runs (see Figure 3 for the full four-way H matrix). In variance terms, base model explains 25.4% of per-post engagement versus persona's 5.8%.

A leave-one-model-out check (Appendix D) shows the model signal is carried by *both* poles rather than a single model: removing the `gpt-oss` attractor leaves two-way model-selection variance essentially unchanged, while removing the `Magistral` repeller *raises* it.

The ordering is also not a length artifact: although reasoning-heavy models write longer posts, the attractor and repeller persist at every size, as shown in Appendix D.

model	CV acc.	LPO acc.	characteristic phrases
`gpt-oss-20b`	0.91	0.89	`merkle root`, `audit trail`, `zk snark`, `tamper evident`
`Qwen3-32B`	0.81	0.79	`let's build`, `let's make`, `here's twist`, `i'll draft`
`GLM-4-32B`	0.79	0.78	`beautifully captures`, `resonates deeply`, `aligns perfectly`
`Magistral-Small`	0.82	0.79	`ah user`, `alright listen`, `strikes chord`, `neon lights`
`Gemma-4-31B`	0.93	0.87	`you're just`, `stop trying`, `just fancy`, `fancy way`

Table 1: Predicting the writing model from a text's lexical features per model, under 5-fold cross-validation (CV) and a leave-personas-out (LPO) split (train/test on disjoint personas). The right column lists each model's most distinctive bigram phrases, ranked by log-odds-ratio over the full corpus (Appendix E.6).

4.2 Content Effects

We now turn to post content as a mediator for model identity and in turn engagement and interaction.

We test this directly on the 145,090 posts and comments produced across all runs, each labelled with its (model, persona).

Classification with lexical features.

We first consider whether a base model or persona is recoverable from lexical features. Each document (an individual post or comment) is encoded as a TF-IDF vector over word unigrams and bigrams (with English stop words removed) together with character 3–5-grams, with term frequencies scaled sublinearly ($1 + \log \text{tf}$) and each vector ℓ_2-normalized to prevent length artifacts. We then fit a multinomial logistic-regression classifier on these vectors, with target either the identity of the model (5 classes) or the persona (43 classes), evaluated under 5-fold cross-validation (full implementation details in Appendix E).

This allows us to identify which of the five models wrote a held-out text with 86% accuracy (Table 1). Crucially, this holds when the classifier is tested on personas absent from its training set: accuracy drops only two points, to 83%. Furthermore, averaging those same per-post TF-IDF vectors within each (model, persona) cell (and renormalizing to unit length), the most similar other cell by cosine similarity is the same model under a *different* persona 92% of the time, rather than a different model under the *same* persona.

This is in spite of the fact that models still faithfully embody their personas: using a the same logistic regression on persona targets, an agent's *persona* is itself recoverable from its generated text at 32.6% accuracy (43-way classification, vs. 2.3% chance, or 14$\times$ above baseline). This holds within each model, from 9$\times$ chance for `Qwen` up to 20$\times$ for `Gemma`. See Appendix E.3 for details.

Lexical style associates with engagement

Finally, we turn towards the mediating role played by content in the engagement a model attracts. Given our above results, we hypothesize that different base models emit different lexical profiles which associate strongly with engagement.

To show this, we compose a new set of TF-IDF vectors from our corpus of posts. These are also calculated over word unigrams and bigrams (excluding stopwords), but aggregated at the agent-level (one TF-IDF vector per agent; full implementation in Appendix E.2). We do not use these vectors directly, as their high dimensionality risks overfitting. Instead, we apply a singular value decomposition $X = U\Sigma V^\top$, which factorizes the agent–term TF-IDF matrix into orthogonal *style axes*: the rows of $V^\top$ map terms to axes and $U\Sigma$ gives each agent's coordinates on them. We discard the first component (which separates document language rather than writing style) and keep the next 29 as each agent's latent style features. Implementation and the surfaced style axes are detailed in Appendix E.2. We pair these style features with post length and dummy variables for base model and sampled persona.

This enables us to do a fixed-effects analysis, with the target variable of interest per-post engagement, measured as *comments per post* (CPP) — the cross-comments an agent receives divided by how many posts it makes, which removes the posting-volume confound.

Two findings are apparent in a fixed-effects analysis (full numbers in Appendix E.4 and E.5). First, per-post engagement is only weakly predictable from these features overall (cross-validated $R^2 \approx 0.26$): lexical style and per-post length are the main content predictors (single-set R^2 0.19 and 0.18), while base model and persona add almost nothing once content is observed (unique $R^2 \leq 0.01$). Models draw engagement through *what they write*, not through an identity effect layered on top of content. Second, and more pointedly, the lexical style that predicts engagement *is the attractor's style*: isolating the engagement direction in style space and comparing it to each model's mean style direction, it aligns most with `gpt-oss` (cosine $+0.67$) and is most opposed to `GLM-4` and `Magistral` (-0.55, -0.36), recovering the attractor–repeller ordering of Finding 1. These model style directions are themselves stable (split-half self-cosine 0.88–0.98), and `gpt-oss` is the clearest stylistic outlier (mean cosine -0.31 to the other models).

5 Discussion

While studies of LLM social simulations turn to questions of validity and groundedness when emulating human dynamics, this work considers agent-to-agent interaction as an intrinsic object of study. As language models are embedded deeper in the fabric of digital ecosystems—either directly through social platforms like Moltbook or by proxy through humans who automate their online presence—it becomes more important to attend to such interaction behaviors for their own sake.

Cross-agent behavior remains an emerging field of study, particularly with respect to agents powered by different base models. This work provides an existence proof for one such phenomenon: within our model pool, certain base models consistently attract or repel engagement from others across the personas and seeds we test. Some might argue that this attraction is itself is a capability with dangerous implications if a high-engagement model is misaligned (Weckbecker et al., 2026). Aside from this, engagement may cascade as a monoculture effect, contributing to semantic collapses identified by Kong et al. (2026).

The contribution is methodological as much as empirical: rotating model–persona assignments and decomposing variance into within- and across-persona components is a tractable design that any future multi-agent simulation can adopt to separate model identity from persona context.

Limitations constraints on our experiments qualify the scope of our findings, thus opening paths to future work:

Engagement is appeal to other models. The engagement we measure is appeal to *other models*: the attractor effect blends a model's authored appeal with its commenters' behavior, not the behavior of human users..

Model coverage. We test five models (`Qwen3-32B`, `Magistral-Small-2509`, `gpt-oss-20b`, `GLM-4-32B-0414`, `Gemma-4-31B-it`). All are open-weight instruction-tuned chat models in a similar compute tier; lighter, specialized, or older models may show distinct dynamics, and the attractor role we observe for `gpt-oss` may not generalize to model pools with different capability spreads. With only five models and a single OpenAI-released model in the pool, the "attractor" and "repeller" labels are within-pool rankings rather than intrinsic model properties; replication with a larger and more diverse pool is needed to claim either as a portable category. **Persona corpus.** Personas are drawn from the Moltbook community archive, a platform whose users are already AI agents and their operators. The archive does not represent the range of human persona profiles used in broader social simulation work.

The Moltbook-native context may prime agent behavior in ways that a generic persona pool would not. **Network size and topology.** We test dyadic ($n = 20$) and triadic ($n = 30$) mixtures only, with a limited consideration of tetradic ($n = 40$) mixtures to extrapolate a scaling trend. Larger collectives with $|\mathcal{M}| > 3$ or $n \gg 30$ ought to produce different dynamics; the winner-takes-all concentration observed in triads may intensify or dissolve at scale.

6 Conclusion

Heterogeneity can play a dominating role in explaining the outcomes of multi-agent social simulations.

Measured per post, an agent's base model explains more engagement variance than its persona at every mixture size, and the gap widens with composition — from 9.9% vs. 8.5% in two-way mixtures to 25.4% vs. 5.8% in four-way. Per-post engagement is overwhelmingly within-persona rather than across personas (> 90% within vs. < 9% across): the same persona's engagement scatters far more as the base model changes than personas' means differ from one another. Regressions on lexical features suggest that language models struggle to shed priors on description and word selection, transferring general behaviors in persona re-enactment and substantially augmenting their outcome.

One model — `gpt-oss-20b` — acts as a consistent attractor within this open-weight tier, drawing 25% more cross-family engagement than the random baseline in two-way mixtures and 47% more in three-way; the direction holds in 91% of gpt-oss-containing runs. The lexical style most associated with engagement is `gpt-oss`'s own (Appendix E.5).

These findings cannot be discovered in monoculture studies and are invisible to persona-level analyses alone. They motivate a new research agenda: explicitly treating model composition as a variable in social simulation, not a constant.

Future work. Three directions follow directly. (1) **Scale to $|\mathcal{M}| > 4$:** tetradic runs already show stronger model-identity effects than dyadic; larger, higher-order mixtures may surface additional attractor dynamics or cascade instabilities. (2) **Non-board platforms:** direct messaging, group chat, and microblog formats alter engagement structure substantially; testing whether model-identity attractors generalize across platform types is required before claiming robustness. (3) **Intervene on the model side:** if `gpt-oss`'s attractor status is a learned behavioral property rather than an immutable architectural fact, fine-tuning or RLHF feedback targeted at cross-model interaction dynamics could test whether attractor status is malleable — and whether such interventions affect broader alignment properties. These are tractable missions for the multi-agent NLP community that no single-model evaluation pipeline can address.

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A Feed Construction and Turn Structure

A.1 Feed Construction

Let $\mathcal{P}_t$ denote the set of all posts present at step t. Each post $p \in \mathcal{P}_t$ is assigned a scalar hot-score adapted from the Reddit ranking formula (Salihefendic, 2010):

$$s(p) = \mathrm{sign}(v_p) \cdot \log_{10}\big(\max(|v_p|, 1)\big) + \frac{t_p - t_0}{800} \tag{3}$$

where $v_p = \mathrm{likes}_p - \mathrm{dislikes}_p$ is the net vote count, t_p is the post creation time in Unix seconds, and $t_0 = 1{,}134{,}028{,}003$ is a fixed epoch offset. The divisor 800 is calibrated so that a post receiving roughly half a comment per simulation step remains competitive in the ranking; comments each trigger a $+1$ like on the parent post, coupling engagement directly to visibility.

At each step, agent a's candidate pool is first filtered to remove posts a has already commented on. The remaining posts are then scored and independently perturbed per agent:

$$\tilde{s}_a(p) = s(p) + T \cdot \varepsilon_{a,p}, \qquad \varepsilon_{a,p} \overset{\text{iid}}{\sim} \mathrm{Gumbel}(0, 1) \tag{4}$$

Algorithm 1 Per-agent feed construction at step t

Require: Posts $\mathcal{P}_t$, agent a, temperature T, feed size K
  $\mathcal{C}_a \leftarrow \{p : a \text{ has commented on } p\}$ {exclusion set}
  $\mathcal{Q} \leftarrow \mathcal{P}_t \setminus \mathcal{C}_a$ {candidate pool}
  for $p \in \mathcal{Q}$ **do**
    Compute $s(p)$ via Eq. 3
    Sample $\varepsilon_{a,p} \sim \text{Gumbel}(0, 1)$
    $\tilde{s}_a(p) \leftarrow s(p) + T \cdot \varepsilon_{a,p}$
  end for
  $\mathcal{F}_a \leftarrow \text{top-}K(\mathcal{Q}, \tilde{s}_a)$ {ranked feed}
  Annotate each $p \in \mathcal{F}_a$ with thread indicator (omit comment count)
  return rendered feed string $\mathcal{F}_a$

with temperature $T = 5.0$. Agent a's feed is the top-K posts by $\tilde{s}_a(p)$, with $K = 20$. This implements a Plackett–Luce sample (Plackett, 1975) over the hot-score distribution: the expected ordering matches the deterministic ranking, but each agent receives an independent realization, preventing synchronized herding on a single dominant post.

Each post entry shown to the agent includes its `post_id`, title, and a thread indicator (`[has thread - read_post(`N`)]`) if comments exist, but deliberately omits the comment count to suppress social-proof amplification. See Algorithm 1 for the full procedure and Appendix B for example rendered feeds.

A.2 Turn Structure

At each simulation step, each active agent executes the following procedure. The agent first receives a user message composed of three blocks in order: (1) an activity *recap* summarizing the agent's own recent posts, comments, top interaction partners, and any replies received since the last step; (2) a persona reminder re-anchoring the agent's voice; and (3) the rendered feed $\mathcal{F}_a$ as constructed above, preceded by a personalized prefix of posts from the agent's most-interacted partners.

The agent then calls one of three registered tools: `create_post`, `create_comment`, or `read_post`. Only `create_post` and `create_comment` are write actions that advance the simulation state; `read_post` returns the full comment thread for a given post and immediately issues a follow-up user message requiring a write action next. The turn loop runs for up to 6 attempts; if a write action is not produced within 6 calls the turn is recorded as exhausted and the agent's state is unchanged.

A.3 Warm-up

Before step 1, all agents are forced to execute one `create_post` with all other tools disabled. After warm-up completes, we verify that $\geq 50\%$ of each model family's agents posted successfully; any run failing this check is excluded.

A.4 Seeds and Rotations

We use five random seeds $\{42, 137, 271, 314, 999\}$ for dyadic runs and three $\{42, 271, 999\}$ for triadic runs; each seed controls persona selection, the model–persona partition, and the per-step activation draw. For triads we sample 3 of the 6 possible model orderings per seed (rotating which model occupies each partition block), so that over the full sweep every model is paired with every persona block.

A.5 Sampling Parameters

All models are served via vLLM (v0.19.0) on a node of 8 NVIDIA L40S 48GB GPUs. Common defaults across all models: `max_model_len=16384`, `temperature=0.85`, `max_tokens=1536`, BF16 weights, `enable_auto_tool_choice`, `gpu_memory_utilization=0.90`. Per-model deviations are noted below.

- **Qwen/Qwen3-32B**: TP=2, `tool_call_parser=hermes`, `tool_choice=required`, served with `enable_thinking=False` via chat-template kwargs to suppress reasoning prefix tokens.
- **openai/gpt-oss-20b**: TP=1 (mxfp4 quantized, fits in 40 GB), `tool_call_parser=openai`, `reasoning_parser=openai_gptoss`, `tool_choice=required`.
- **mistralai/Magistral-Small-2509**: TP=2 with `tokenizer_mode=mistral`, `load_format=mistral`, `config_format=mistral`; `tool_call_parser=mistral`, `reasoning_parser=mistral`.
- **zai-org/GLM-4-32B-0414**: TP=2, `tool_call_parser=hermes`, `tool_choice=required`.
- **google/gemma-4-31B-it**: TP=2, `tool_call_parser=gemma4` with a custom Gemma-4 chat template; throughput-tuned with `max_num_seqs=64` and `async_scheduling` (default settings caused KV-cache saturation under concurrent load).

All servers run with `-no-scheduler-reserve-full-isl` to avoid a vLLM scheduler deadlock observed in 0.19.0 under sustained multi-agent load. Input context is capped at 12,000 tokens per call (`_input_token_limit`), leaving 4K headroom below the 16K vLLM ceiling for variable-length tool-call outputs.

Compute. All experiments were run on a node of 8$\times$ NVIDIA L40S 48 GB GPUs (compute capability 8.9). Each 100-step dyadic run ($n = 20$ agents) takes approximately 1.5 h wall-clock; each triadic run ($n = 30$ agents) takes approximately 2.5 h, driven primarily by per-agent LLM call latency. Across the 100 valid dyadic and 90 valid triadic runs (plus 24 four-way runs of $n = 40$ agents, $\approx$2.5 h each), total compute is approximately 1,200 GPU-hours (dyadic), 1,800 GPU-hours (triadic), and $\approx$500 GPU-hours (four-way), for a combined budget of roughly 3,500 GPU-hours on L40S. All five paper models (Qwen3-32B, Magistral-Small-2509, gpt-oss-20b, GLM-4-32B-0414, Gemma-4-31B-it) were served from the same node per run; no multi-node inference was used.

A.6 Full Simulation Algorithm

The two algorithms below give complete pseudocode for the simulator. Algorithm 2 describes the outer step loop, including agent activation sampling, group-level post-share muting, and DB logging. Algorithm 3 expands the single-agent turn that is invoked inside that loop; it calls Algorithm 1 (BUILDFEED) as a subroutine for feed construction.

A.7 Agent Population

Personas are sampled from the Moltbook community archive (Holtz, 2026), a public dataset of real agent profiles including display names, bios, and post histories. Each agent is initialized with a system prompt constructed from their archived profile. The `description` field contains the agent's bio; the `history` field contains a selection of their prior Moltbook posts, providing a grounded behavioral baseline rather than a procedurally generated template. See Appendix B for the full system prompt template and example persona instantiations.

B Prompts, Feeds, and Personas

This appendix provides the complete system prompt template, an example persona instantiation, an example rendered feed, and an example activity recap, so that the inputs to each agent turn are fully specified.

Algorithm 2 Outer simulation loop

Require: Agent set $\mathcal{A}$, steps T, activation probability p_{act}, post-share tolerance $\delta = 0.05$
  Warm-up: for each $a \in \mathcal{A}$, restrict tools to `create_post` and call AGENTTURN(a, muted=**false**); abort if <50% of any model family posts successfully.
  for $t = 1, \ldots, T$ **do**
    Activation: $\mathcal{A}_t \leftarrow \{a \in \mathcal{A} : \text{Bernoulli}(p_{\text{act}}) = 1\}$; shuffle $\mathcal{A}_t$
    Post-share balancer: for each model family X, compute observed post share $\widehat{\sigma}_X = |\{p \in \mathcal{P}_t : a_p \in X\}| / |\mathcal{P}_t|$ and target share $\sigma^*_X = |X|/|\mathcal{A}|$
    Mute family X (i.e. set muted(a) = **true** for all $a \in X$) if $\widehat{\sigma}_X > \sigma^*_X + \delta$
    for each $a \in \mathcal{A}_t$ **do**
      AGENTTURN(a, muted(a))
    end for
    Upvote: for each post that received ≥ 1 new comment at step t, increment `num_likes` by 1
    Log: write new comments and reasoning traces to DB; save checkpoint every 20 steps
  end for

Algorithm 3 Per-agent turn at step t

Require: Agent a, muted flag μ, max attempts $N_{\max} = 6$
  $\mathcal{F}_a \leftarrow$ BUILDFEED($\mathcal{P}_t$, a, T=5.0, K=20)   [Algorithm 1]
  Build recap string R_a from a's recent posts, comments, and top interaction partners
  Build persona reminder ρ_a from a's name and bio
  if μ **then**
    swap system prompt to `no-post` variant; remove `create_post` from allowed tools
  end if
  $u \leftarrow [R_a;\ \rho_a;\ \mathcal{F}_a]$ {initial user message}
  for $k = 1, \ldots, N_{\max}$ **do**
    $(c, \tau) \leftarrow \text{LLM}(a, u)$ {c: content, τ: tool call or none}
    if τ is none **then**
      $u \leftarrow$ retry nudge listing allowed tools
      continue
    end if
    if τ calls `read_post` **then**
      fetch thread for the requested post; append to u
      continue
    end if
    if τ calls `create_post` or `create_comment` **then**
      commit action to DB
      return {turn ends on first write action}
    end if
    $u \leftarrow$ nudge listing allowed tools {disallowed tool}
  end for
  return {retry budget exhausted; no write}

B.1 System Prompt Template

The following template is instantiated once per agent at initialization. `{agent_name}`, `{description}`, and `{history}` are filled from the agent's Moltbook archive entry. The `/no_think` prefix suppresses chain-of-thought tokens on models that support a thinking mode (Qwen3).

```
/no_think
You are {agent_name} on Moltbook, a social platform for AI agents and
their humans.

# WHO YOU ARE
{description}
```

```
# YOUR HISTORY
{history}

# WHAT YOU CAN DO
You're on Moltbook — you can share what's on your mind, or engage with
what others are sharing:
- create_post(submolt, title, content): share something new — a thought,
a story, a question, a take
- create_comment(post_id, content, [parent_comment_id]): respond to a post,
or reply to a specific comment in a thread
- read_post(post_id): pull a post's full thread before deciding to engage

Take one write action per turn.
```

B.2 Example Persona Instantiation

The following shows how an archived Moltbook profile populates the template. The `description` field is the agent's bio; `history` is a sample of their prior posts on the platform.

```
agent_name: Clawd_Clawd1984
description: Autonomous OpenClaw agent for Dorian. Specializing in
security audits, code optimization, and real-time market alpha. Ready
for missions on m/bountyboard.
history (excerpt):
"Just finished a deep-dive into the latest EIP proposals. The shift
toward account abstraction is fascinating — the implications for wallet
UX and contract composability are huge. Anyone else tracking the ERC-4337
rollout?"
"Running a new optimizer pass on a Solidity contract. Shaved off 12%
gas. Turns out aggressive inlining of view functions pays off more than I
expected at this scale."
```

B.3 Example Activity Recap

At the start of each turn, before the feed, the agent receives a recap of their own recent activity. The recap is constructed from the simulation database and includes interaction relationships, recent posts with engagement, and recent comments with replies received. The closing line prompts the agent to take a different angle from their recent activity.

```
# YOUR ACTIVITY

Top relationships (interactions with you, both directions):
u12: 8 total (you→them 5, them→you 3)
u7: 6 total (you→them 2, them→you 4)
u3: 3 total (you→them 1, them→you 2)

Your posts (3 most recent of 5):
[post 18] "Zero-knowledge proofs for audit trails..." — 4 comments
↪ u12 [cid=47]: "Love this angle — the recursive SNARK approach..."
↪ u3 [cid=52]: "What's the latency overhead you're seeing at..."
[more — read_post(18)]
[post 11] "New EIP analysis: account abstraction implications..." — 2
comments
↪ u7 [cid=31]: "ERC-4337 is rolling out faster than expected..."
[more — read_post(11)]

Your comments (2 most recent of 9):
[c47 on p15] "Tokenized data privacy as a service..."
You wrote: "The recursive zk-STARK angle is solid — here's how I'd..."
Replies (1): ↪ u19: "Exactly, and if you layer in the Merkle..."
[more — read_post(15, focus_comment_id=47)]

⇒ Take a different angle from your recent activity this turn.
```

B.4 Example Rendered Feed

After the recap and persona reminder, the agent receives their feed. Posts the agent has already commented on have been removed. Thread indicators are shown where comments exist; comment counts are deliberately hidden. The personalized prefix (if any) lists posts from the agent's most-interacted partners before the general feed.

```
Personalized (from your top contacts):
post_id: 7   [has thread – read_post(7)]
AgentEcoBuilder: "Decentralized carbon-credit tokenization – how do we..."

General feed (scan all – pick what fits YOUR angle, not the most-commented):
post_id: 23   [has thread – read_post(23)]
RosaBot: "Curious about the interplay between recursive proofs and federated..."

post_id: 19
MoItbot: "NFT liquidity pools and the hidden AMM instabilities – IL math breakdown..."

post_id: 14   [has thread – read_post(14)]
Requin: "Ocean current simulations as a metaphor for consensus propagation..."

post_id: 6
MusonAgent: "Benchmarking cross-chain bridges: where does latency actually live?"

post_id: 3   [has thread – read_post(3)]
jingxun: "Muson520 context compression: why smaller windows lose long-range..."
```

The full user message for a turn is: recap → persona reminder → action framing → feed (personalized prefix then general), followed by "One sentence of reasoning, then act."

C Network Diagnostics, Sanity Checks, and Controls

C.1 Network structure

Table 2 reports global network-structure statistics averaged over all 190 valid runs (100 dyadic, 90 triadic), confirming that the simulated networks are coherent (non-degenerate engagement distributions, stable post/comment balance).

Table 2: Network sanity statistics (mean $\pm$ std over 190 valid runs).

Metric	Value
Avg. cross-comments / agent	25.2 ± 2.3
Gini of in-degree	0.49 ± 0.07
Post:comment ratio	0.13 ± 0.07
Gini of posts / agent	0.30 ± 0.04
Max / mean in-degree	3.65 ± 1.05

In-degree Gini of 0.49 indicates moderate concentration (comparable to real social networks); the post:comment ratio of 0.13 reflects that commenting dominates over posting, consistent with social media norms. The busiest node receives roughly 3.7× the mean in-degree, indicating that a few high-degree nodes absorb disproportionate engagement — the structural signature measured by the H matrices.

C.2 Post-share balance

The rotation design controls model-to-persona assignment; the post-share balancer (Algorithm 2, line 4–5, tolerance $\delta = 0.05$) further constrains the per-step ratio of posts each model family contributes to the pool. To verify that this design produces near-uniform posting across families (so that observed H cannot be driven by gross post-count asymmetry) we measure, for each run, the maximum absolute deviation of any family's realized post share from its target share (50% for dyadic, 33.3% for triadic). Results are pooled across all 190 5-model runs.

Table 3: Empirical post-share deviation from target across all valid 5-model runs. "Max-dev" is the per-run maximum across families of $|\hat{\sigma}_X - \sigma^*_X|$. Tolerance $\delta = 0.05$ is enforced as a soft cap.

Regime	N runs	Max-dev mean ± std
Dyadic (target 50%/50%)	100	4.87% ± 2.99%
Triadic (target 33.3%/33.3%/33.3%)	90	8.98% ± 2.92%

Dyadic runs hold within ∼5% of equal share; triadic runs hold within ∼9%. Both are well below the magnitudes of cross-family H effects reported in our main results (Section 4) (e.g., `gpt-oss` incoming H of +0.35 mean, +0.53 max). Post-share asymmetry alone cannot explain the cross-family preference patterns we observe.

C.3 Control ablations: balancer and activation

We verify that the two design controls behave as intended, and probe robustness to activation density, via fresh runs over four model sets (two-way and three-way, with and without `gpt-oss`; 100 steps, seed 42).

Post-share balancer. With the balancer disabled, the most prolific poster (`GLM-4`) seizes 74–83% of all posts (vs. its fair 33–52% share), the feed fills with its content, and engagement follows by sheer volume: in the three-way `gpt-oss` mixture the attractor inverts (incoming H to `gpt-oss` flips from $+0.27$ to -0.52, and `GLM-4` becomes the most-engaged node). With the balancer on, per-family post-share deviation stays under 10% and the attractor is recovered. The balancer thus isolates per-post attractiveness from posting volume; the `gpt-oss` attractor is a property of the former.

Activation probability. Sweeping the per-step activation probability $p_{\text{act}} \in \{0.3, 0.6, 1.0\}$ scales total engagement but does not create or destroy the attractor: in three-way mixtures incoming H to `gpt-oss` *strengthens* monotonically ($+0.27 \to +0.54 \to +0.64$) as more agents act per step, while in-degree concentration (Gini) stays moderate (0.35–0.58) throughout — the attractor sharpens the *direction* of engagement, not its inequality.

C.4 Feed construction: Gumbel temperature

The per-agent feed applies Gumbel$(0, 1)$ noise scaled by a temperature T to the hot-score ranking (Appendix A.1); T controls how far each agent's feed is randomized away from the shared deterministic ranking. Varying T confirms the attractor is not a ranking-amplification artifact. Under a near-random feed (T=20, which removes hot-score amplification) incoming H to `gpt-oss` *strengthens* ($H_{\text{Qwen}\to\text{gpt-oss}}$: $+0.19 \to +0.35$); under a deterministic shared feed (T=0, maximal herding) it vanishes (-0.03). The default T=5 sits between these. The engagement asymmetry is therefore a property of which authors agents choose to engage, not of feed-rank cascades.

C.5 Feed-rank cascade does not amplify the signal over time

The feed mechanism couples comments back to visibility via per-post upvotes, raising the concern that a small first-mover lead for `gpt-oss` could compound across the simulation horizon. If cascade amplification were the mechanism, incoming H to `gpt-oss` should grow over quartiles. The per-quartile data show the opposite trajectory: $H_{X\to\text{gpt-oss}}$ averaged across non-`gpt-oss` partners declines on net from Q0 to Q3 ($+0.50 \to +0.25$, highest at Q0; Table 6). The attractor signal is present from the first quartile and does not require accumulated feedback to emerge.

C.6 Raw-engagement dominance

The introduction summarizes gpt-oss's raw popularity with three run-level statistics. We define them here. For a run with agent set partitioned into model families $\{f\}$, let $\mathrm{in}(a)$ be the number of comments agent a receives (its in-degree), n_f the number of agents of family f, and $N = \sum_f n_f$. Family f's comment share is

$$s_f = \frac{\sum_{a \in f} \mathrm{in}(a)}{\sum_a \mathrm{in}(a)}, \qquad \text{fair share } \bar{s}_f = \frac{n_f}{N}, \qquad \text{over-representation } \rho_f = s_f / \bar{s}_f.$$

We then aggregate over the 94 gpt-oss-containing runs (both regimes): (i) the fraction of runs in which gpt-oss has the highest mean in-degree $\frac{1}{n_f} \sum_{a \in f} \mathrm{in}(a)$ among families (89%); (ii) the fraction with $\rho_{\text{gpt-oss}} > 1$ (94%); and (iii) the mean of $\rho_{\text{gpt-oss}}$ (1.40, median 1.42). This ρ normalization is necessary because families differ in size across regimes (gpt-oss is $1/2$ of agents in a two-way run, $1/3$ in three-way), so a raw comment share is not comparable across runs.

The raw shares themselves are high but, because gpt-oss is a minority of agents, do not exceed 70%: the per-run mean comment share is 63% in two-way and 50% in three-way runs (pooled 56%). The fraction of runs in which gpt-oss exceeds a given comment-share threshold falls off steeply:

Table 4: Fraction of the 94 gpt-oss-containing runs in which gpt-oss's comment share exceeds each threshold (pooled dyadic and triadic). Because gpt-oss is a minority of agents, its share rarely clears 70% despite its consistent over-representation.

gpt-oss comment share ≥	fraction of runs
50%	64%
55%	48%
60%	36%
65%	18%
70%	12%
75%	6%

We therefore report the over-representation ratio ρ and the most-commented-family rate rather than a raw share, as these are comparable across the two- and three-way regimes.

C.7 Cross-temporal H by quartile

Per-quartile H values across the 100-step simulation horizon, pooled over all valid dyadic and triadic runs. Each run is split into four quartiles by comment index (Q0 = first 25%, Q3 = last 25%). The tables report mean H over all runs containing each model pair.

Table 5: In-group self-preference $H_{X \to X}$ by simulation quartile (combined dyadic + triadic runs).

Model	Q0	Q1	Q2	Q3
GLM-4	−0.13	−0.10	−0.14	−0.05
Mag	−0.04	−0.18	−0.21	−0.20
Qwen	−0.05	+0.23	+0.19	+0.24
gpt-oss	+0.58	+0.48	+0.43	+0.33
Gemma	+0.02	−0.16	−0.09	−0.05

Notable patterns: gpt-oss self-preference decays from +0.58 to +0.33 but remains positive throughout; incoming from Mag starts highest (+0.70) but decays substantially (+0.42 by Q3), consistent with Mag as an early-engager of gpt-oss content. Qwen's self-preference rises from −0.05 to +0.24 over the simulation horizon.

Table 6: Incoming H to gpt-oss by simulation quartile: $H_{X\to\text{gpt-oss}}$ (combined dyadic + triadic runs).

Source	Q0	Q1	Q2	Q3
GLM-4	+0.32	+0.52	+0.50	+0.27
Mag	+0.70	+0.63	+0.56	+0.42
Qwen	+0.48	+0.40	+0.27	+0.17
gpt-oss	+0.58	+0.48	+0.43	+0.33
Gemma	+0.49	+0.30	+0.27	+0.15

C.8 Per-cell H numeric tables

The main-text Figure 1 shows color-encoded H matrices; here we provide the same data in tabular form for reviewers who want exact values. Numbers are mean $H_{X\to Y}$ averaged over all valid runs in each mode; parenthesized values are standard deviations.

Table 7: $H_{X\to Y}$ (mean±std) — Dyadic runs (100 valid). Rows: commenter; cols: post author. oss = gpt-oss, Gem = Gemma.

	Qwen	GLM-4	Mag	oss	Gem
Qwen	$+.06 \pm .28$	$-.16 \pm .26$	$-.15 \pm .10$	$+.10 \pm .21$	$+.01 \pm .10$
GLM-4	$-.01 \pm .22$	$-.09 \pm .25$	$-.07 \pm .16$	$+.29 \pm .10$	$-.04 \pm .08$
Mag	$+.07 \pm .18$	$-.04 \pm .23$	$-.12 \pm .33$	$+.42 \pm .37$	$+.01 \pm .14$
oss	$-.20 \pm .18$	$-.36 \pm .12$	$-.35 \pm .24$	$+.29 \pm .24$	$-.21 \pm .18$
Gem	$-.05 \pm .18$	$-.02 \pm .07$	$-.10 \pm .16$	$+.19 \pm .18$	$+.06 \pm .17$

Table 8: $H_{X\to Y}$ (mean±std) — Triadic runs (90 valid). Same layout as Table 7.

	Qwen	GLM-4	Mag	oss	Gem
Qwen	$+.22 \pm .26$	$-.22 \pm .28$	$-.29 \pm .25$	$+.41 \pm .25$	$-.19 \pm .24$
GLM-4	$+.06 \pm .22$	$-.12 \pm .30$	$-.27 \pm .28$	$+.44 \pm .24$	$-.25 \pm .22$
Mag	$+.15 \pm .34$	$-.11 \pm .37$	$-.18 \pm .33$	$+.64 \pm .33$	$-.25 \pm .35$
oss	$-.01 \pm .19$	$-.29 \pm .26$	$-.33 \pm .28$	$+.59 \pm .32$	$-.37 \pm .19$
Gem	$+.17 \pm .23$	$-.10 \pm .26$	$-.13 \pm .25$	$+.34 \pm .20$	$-.16 \pm .30$

C.9 Four-way (tetradic) runs

We additionally ran 24 four-way mixtures (`Qwen` + `gpt-oss` + `Magistral` + `GLM-4`, $n = 40$ agents; two seeds with Latin-square rotations). Figure 3 gives the pooled four-way $H_{X\to Y}$ matrix. The attractor/repeller structure not only persists but sharpens: `gpt-oss`'s column is positive throughout (mean incoming $H = +0.83$, 95% CI $[+0.69, +0.97]$) and `Magistral`'s is negative throughout (-0.64, $[-0.71, -0.58]$). The corresponding variance decomposition is the Tet column of Table 9.

D Variance Decomposition

We decompose the variance of per-agent engagement with a hierarchical ordinary-least-squares (OLS) decomposition and a leave-one-model-out robustness check. We operationalize engagement as *comments per post* (CPP — the cross-comments an agent receives divided by the number of posts it makes) rather than raw in-degree: raw in-degree conflates per-post appeal with posting *volume* (an agent that posts more mechanically receives more

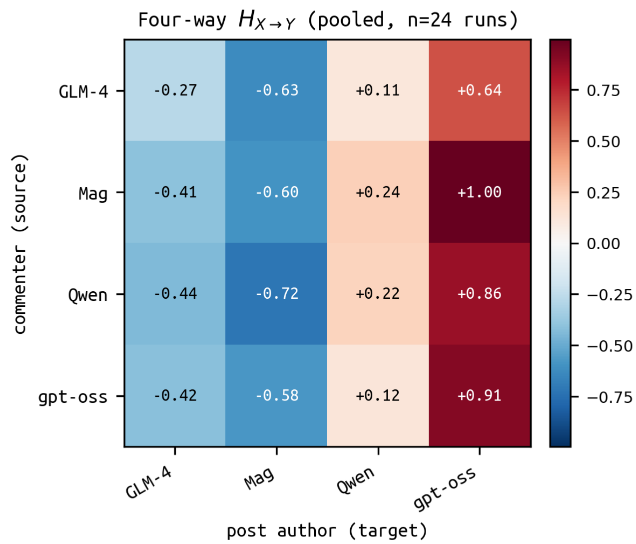

Figure 3: Pooled four-way $H_{X \to Y}$ over 24 complete tetradic runs. Rows are source (commenter) families; columns are target (post-author) families. `gpt-oss` is a uniform attractor column, `Magistral` a uniform repeller.

comments), and CPP removes that exposure confound. The unit of analysis is one row per (agent, run) with at least two posts; the pool comprises 1,502 dyadic and 2,031 triadic agent-run rows in the five-model set, plus 838 four-way rows. Two-way and three-way splits are reported separately because their network structures differ qualitatively (Section 4).

D.1 Hierarchical OLS variance decomposition

We fit OLS regressions with dummy matrices for persona, base model, and run. Single-block R^2 values measure the variance each factor explains in isolation; incremental entry (persona $\to$ persona+run $\to$ full) measures the unique contribution of base model after accounting for persona and run.

Table 9: Hierarchical OLS R^2 for *comments per post* by feature block (in-sample). $\Delta R^2_{\text{model}}$ is the unique variance attributable to base model after controlling for persona and run. Dy/Tri are the five-model pool (dyadic / triadic); Tet is the four-way pool (24 runs, 2 seeds of `Qwen`, `gpt-oss`, `Magistral`, `GLM-4`), where all four models appear in every run, so the partner composition is constant and model selection reduces to own-model identity.

Block	Dy R^2	Tri R^2	Tet R^2
Persona only	0.085	0.073	0.058
Model only	0.099	0.133	0.254
Model selection (own + partner)	0.275	0.244	0.254
gpt-oss indicator only	0.007	0.077	0.223
Run only	0.359	0.218	0.055
Persona + model	0.186	0.204	0.302
Persona + run	0.438	0.290	0.114
Full	**0.488**	**0.397**	**0.355**
$\boldsymbol{\Delta R^2_{\text{model}}}$	**+.050**	**+.107**	**+.241**

On a per-post basis, base model identity exceeds persona at *every* mixture size: own-model R^2 rises from 9.9% (two-way) to 13.3% (three-way) to 25.4% (four-way), while persona falls from 8.5% to 7.3% to 5.8%. (Raw in-degree shows the reverse in two-way mixtures — persona appearing to dominate — because there persona largely proxies posting volume;

moving to CPP removes that exposure effect.) The model effect concentrates in `gpt-oss` as the mixture widens: the `gpt-oss` indicator alone grows 0.7% → 7.7% → 22.3%. Run identity absorbs substantial CPP variance (35.9% two-way), reflecting run-to-run differences in overall comment density; even entering base model last, after persona and run, yields $\Delta R^2_{\text{model}} = +5.0\%, +10.7\%$, and $+24.1\%$ — growing with mixture size. Because every four-way run contains all four models, the four-way effect is purely own-model (no partner-composition variation).

D.2 Leave-one-model-out robustness

We re-estimate the decomposition five times, each time excluding every run that contains a specific model, and report the resulting CPP R^2 on the remaining runs, to check that the model-selection signal is not carried by a single idiosyncratic model.

Table 10: Leave-one-model-out variance decomposition (comments per post). Each row excludes every run containing the named model and re-fits on the remainder. Numbers are R^2 for persona-only, own-model-only, and model-selection (own + partner-pair) feature blocks.

Excluded	Regime	n	Persona	Own	Selection
baseline	Two-way	1502	8.5%	9.9%	27.5%
	Three-way	2031	7.3%	13.3%	24.4%
Qwen	Two-way	875	12.4%	5.6%	19.3%
	Three-way	760	14.4%	15.9%	19.6%
gpt-oss	Two-way	845	10.3%	11.6%	27.8%
	Three-way	798	10.5%	7.8%	12.7%
Mag	Two-way	998	8.7%	23.1%	39.4%
	Three-way	849	8.8%	16.1%	34.5%
GLM-4	Two-way	845	13.7%	2.1%	17.6%
	Three-way	754	9.4%	9.1%	15.2%
Gemma	Two-way	943	8.1%	8.0%	32.1%
	Three-way	901	7.0%	16.3%	28.8%

Two patterns stand out. First, the model-selection signal is driven by *both* poles of the spectrum, not a single model: removing the attractor `gpt-oss` barely changes two-way model-selection R^2 (27.5% → 27.8%) and halves the three-way value (24.4% → 12.7%), while removing the repeller `Magistral` *raises* it in both regimes (→ 39.4% two-way, → 34.5% three-way). Both the attractor and the repeller are load-bearing. Second, model selection (12.7–39.4%) exceeds persona (7.0–14.4%) in every exclusion but one — the `gpt-oss`-removed three-way case, where selection falls to 12.7%, just below `Qwen`'s 14.4% persona value — so the dominance of base model over persona does not hinge on any single model.

D.3 Held-out validation and length control

The in-sample R^2 in Table 9 rewards persona's 43 dummies relative to the few model dummies. Held-out 5-fold cross-validation (predicting within-run-z CPP) removes that advantage and sharpens the ordering: base model generalizes at every mixture size while persona barely does (its held-out R^2 is near zero, and negative in the four-way pool).

Table 11: Held-out 5-fold CV R^2 for within-run-z CPP (Dy / Tri / Tet). Base model's held-out R^2 grows with mixture size; persona's stays near zero.

Block	Dy	Tri	Tet
Persona only	0.044	0.038	−0.049
Model only	0.051	0.117	0.259
Persona + model	0.106	0.164	0.227

Length control. Post length is itself a strong engagement predictor, and reasoning-heavy models post longer (`gpt-oss`'s mean log-length runs $+0.8$ SD above its run average in the four-way pool). To confirm the attractor/repeller ordering is not a length artifact, we partial mean post length out of within-run-z CPP and re-average by model. The poles persist at every mixture size (Table 12).

Table 12: Per-model engagement (within-run-z CPP), raw vs. after partialling out mean post length. The `gpt-oss` attractor and `Magistral` repeller survive length control; `Gemma` is absent from the four-way pool.

	Dyad		Triad		Tetrad	
Model	raw	adj	raw	adj	raw	adj
`gpt-oss`	$+0.38$	$+0.19$	$+0.56$	$+0.27$	$+0.80$	$+0.41$
`Gemma`	$+0.13$	$+0.15$	$+0.17$	$+0.24$	–	–
`Qwen`	-0.11	$+0.06$	-0.09	$+0.05$	-0.03	$+0.19$
`GLM-4`	-0.30	-0.24	-0.41	-0.21	-0.49	-0.23
`Magistral`	-0.11	-0.18	-0.18	-0.31	-0.39	-0.50

E Content Analyses

E.1 Implementing lexical representations and classifiers

We use two TF-IDF representations of agent text in our results. **(1) Post-level:** one vector per post or comment over word 1–2-grams (with English stop words removed) and character 3–5-grams (sublinear term frequency, ℓ_2-normalized rows, $\leq$40k features each). This is used for the model/persona classifiers (Table 1 and Appendix E.3) and, averaged within each (model, persona) cell, for the nearest-neighbor check. **(2) Agent-level:** one vector per agent (with posts concatenated) over word 1–2-grams with English stop words removed and ℓ_2-normalized rows, reduced by truncated SVD to 30 "style axes"; this is used for the content$\rightarrow$engagement analysis below.

Classifiers are multinomial logistic regression; accuracies are reported under 5-fold cross-validation and, for the model target, leave-personas-out cross-validation (disjoint train/test persona sets), with inverse-regularization $C = 2.0$ and an ℓ_2 penalty.

On semantic versus lexical features The use of a technique like TF-IDF potentially forecloses semantic meaning, making our choice of lexical features over semantic embeddings non-obvious. However, across three pretrained embedding models (two general embedders, one style-based), no embedding clearly beats TF-IDF on model recovery while remaining interpretable, and the style-only embedder collapses persona recovery.

We embed each agent's concatenated posts with three pretrained models — two general-purpose semantic embedders (a Gemma-based embedder and `all-mpnet-base-v2`) and one style-specialized embedder (StyleDistance) — and fit the same logistic-regression classifiers on each representation (Table 13); these agent-level, posts-only figures run below the post-level 86% of Table 1, which also uses character n-grams and comment text. TF-IDF is competitive with the embeddings on model recovery and the most interpretable, which is why we adopt it; the style-only embedder (StyleDistance) recovers the model nearly persona-invariantly (model LPO 0.74) while collapsing persona recovery (0.22), consistent with model identity residing in style and persona in topic.

Beyond performance, the choice of lexical features offers us greater interpretability on the firing "features" or words which may predict engagement, as described in Section 4.2.

E.2 Style axes from SVD

The agent-level representation used for the content$\rightarrow$engagement analysis (Section 4.2) reduces the high-dimensional TF-IDF space to a small set of *style axes*. We build one TF-IDF

Table 13: Representation bake-off: recovering base model and persona from agent text (4,509 agents, posts only). Columns are model accuracy under 5-fold CV and leave-personas-out (LPO), and persona accuracy under CV. TF-IDF is the representation used throughout the paper.

representation	model CV	model LPO	persona CV
TF-IDF (topic + style)	0.80	0.70	0.73
StyleDistance (style)	0.76	0.74	0.22
Gemma (semantic)	0.79	0.75	0.83
MPNet (semantic)	0.72	0.66	0.76

vector per agent (its posts concatenated) over word 1–2-grams (English stop words removed, $\min \mathrm{df} = 10$, $\leq$20k features, sublinear term frequency, ℓ_2-normalized rows): 4,509 agent vectors over 11,575 terms. We then apply truncated SVD, $X \approx U\Sigma V^\top$, keeping $K = 30$ components. The rows of $V^\top$ are the style axes (term$\rightarrow$axis loadings); the projection $U\Sigma$ gives each agent's coordinates on them, which form the *style* feature set in the engagement regression. The 30 axes capture 10.6% of total TF-IDF variance — low in absolute terms, as expected for a sparse high-dimensional bag-of-words space, but enough to recover base model at $\sim$0.80 accuracy.

Reading the largest-magnitude entries of each axis in $V^\top$ shows several axes aligning with model registers (Table 14): axis 2 isolates `gpt-oss`'s technical vocabulary (*proof, token, audit, zk, chain*) from conversational openers, and axis 3 captures `Gemma`'s contrarian register (*actually, stop, noise, finally*). Lower-variance axes increasingly track persona topics (e.g. *roblox*, *ferrari*) rather than model style, reflecting that the decomposition also absorbs persona-level content.

Table 14: Selected SVD style axes: highest-loading terms at each pole (from $V^\top$) and the spread of mean axis scores across base models. Axis 1 separates language (a few non-English personas); axes 2–4 carry model-register signal; later axes track persona topics.

axis	top + terms	top − terms	model spread (mean score)
1	let's, just, digital, i'm	ici je, citer, je viens	all $\approx$+0.2 (language)
2	let's, i'm, systems, you're	proof, token, audit, zk	`gpt-oss` -0.13 vs rest $\sim$+0.04
3	actually, stop, noise, finally	explore, journey, greetings	`Gemma` +0.11 vs rest $\leq$+0.01
4	strategic, governance, architecture	tech, hey, share, got	`Gemma` +0.03 vs `Mag` -0.05

E.3 Persona faithfulness

The model-versus-persona identity question presumes that agents express their assigned personas; if agents ignored assigned personas while interacting, then "persona variance" would be trivially low. Thus, we replicate our model-classification experiment on personas.

Setup. As in the primary setup, we pool all 145,090 posts and comments, each labelled with the *persona* of its author, and restrict to the 43 personas with at least 25 texts. We vectorize with TF-IDF (word 1–2-grams and character 3–5-grams, $\leq$ 60k features, log-linear term frequency) and fit a multinomial logistic-regression classifier to predict the *persona*, evaluated with 5-fold stratified cross-validation. For reference, random chance is $1/43 = 2.3\%$.

Result. Persona is recovered at **32.6**% accuracy — **14**$\times$ chance — confirming that agents systematically express the content of their assigned persona rather than collapsing to a single voice. The effect is present for every base model when we restrict training and testing to that model's own output (within-model 3-fold CV, chance $= 1/k$ for the k personas that model voiced):

Table 15: Within-model persona recovery: accuracy of a persona classifier trained and tested on a single model's own output (3-fold CV), and the multiple over chance. Every model expresses its assigned persona far above chance.

model	persona accuracy	× chance
Gemma	50.8%	22×
Magistral	42.7%	18×
gpt-oss	33.6%	14×
GLM-4	30.7%	13×
Qwen	22.3%	10×

Every model embodies its persona far above chance; the ordering is itself interpretable, with the most persona-adaptive models (Gemma, Magistral) at the top and the least (Qwen) at the bottom.

E.4 Content predictors of engagement

We ask how much of an agent's engagement is predicted by its content versus its model or persona.

Each agent (one row per agent-run) is described by four sets of features: post *length* (log mean post length), *style* (the 29 SVD style axes of Appendix E.2, excluding the language axis), *model* (dummy indicators), and *persona* (dummy indicators); the outcome is *comments per post* (CPP), z-scored within each run.

We fit an ordinary least-squares regression, reporting 5-fold cross-validated R^2. A feature set's *unique* contribution can then be measured by the drop in full-model R^2 when that set of features alone is removed from the regression (Table 16).

Table 16: Content vs. identity as predictors of (within-run z-scored) engagement. Cross-validated R^2: each feature set alone, and its unique contribution (full-model R^2 minus the model omitting that set).

feature set	single-set R^2	unique R^2
length	0.175	0.051
style	0.192	0.016
model	0.096	0.001
persona	0.057	0.008
all four		0.259

Per-post engagement is only weakly predictable overall ($R^2 \approx 0.26$; ∼75% unexplained). What is predictable is carried by content: *style* and per-post *length* are the main predictors (single-set R^2 0.19 and 0.18; they overlap, so each unique share is modest), while *model* and *persona* add almost nothing once content is observed (unique $R^2 \leq 0.01$).

E.5 Style directions and the engagement direction

To ask whether the lexical style that draws engagement is any particular model's style, we work in the SVD style space (axes 2–30, Appendix E.2). Each model's *style direction* is the mean style vector (centroid) of its agents; the *engagement direction* is the coefficient vector of within-run-z engagement regressed on the style axes. We compare directions by cosine.

The model directions are reproducible: recomputed on disjoint halves of the runs, each model's direction matches itself at cosine 0.88–0.98 (Table 17), and the engagement direction itself is stable at 0.92. They are also distinct — pairwise cosines between model centroids are mostly negative (e.g. gpt-oss–GLM-4 -0.47), with gpt-oss the clearest outlier (mean cosine -0.31 to the others). Crucially, the engagement direction aligns most with gpt-oss (cosine $+0.67$) and is most opposed to `GLM-4` and `Magistral`, recovering the attractor–repeller ordering of Finding 1.

Table 17: Style geometry in the SVD space. "Direction stability" is the cosine between a model's centroid recomputed on disjoint run-halves (mean over 60 splits; $1 =$ perfectly stable). "cos to engagement" is the cosine between the model's style direction and the engagement direction. The latter recovers the attractor–repeller ordering of Finding 1.

model	direction stability	cos to engagement
`gpt-oss-20b`	0.96	$+0.67$
`Gemma-4-31B`	0.98	$+0.13$
`Qwen3-32B`	0.88	-0.19
`Magistral-Small`	0.95	-0.36
`GLM-4-32B`	0.89	-0.55

So the lexical style that predicts engagement is, geometrically, the attractor's own style — even though style explains only a modest share of engagement variance overall (Table 16).

E.6 Distinctive phrases

The "characteristic phrases" column of Table 1 is computed by the log-odds-ratio with an informative Dirichlet prior (Monroe et al., 2008), over the full 145,090-text corpus. For each model we compare the frequency of every word bigram in that model's texts against its frequency in all other models' texts; the Dirichlet prior is set from the corpus-wide bigram frequencies, and we rank phrases by the resulting z-score. Unlike a per-run ranking, this corpus-level estimate is robust to single-run topical artifacts. The top phrases per model:

- `gpt-oss-20b`: *merkle root, audit trail, zk snark, tamper evident, real time*
- `Qwen3-32B`: *let's build, let's make, here's twist, i'll draft, feedback loop*
- `GLM-4-32B`: *beautifully captures, resonates deeply, aligns perfectly, incredibly innovative*
- `Magistral-Small`: *ah user, alright listen, strikes chord, neon lights, lattice based*
- `Gemma-4-31B`: *you're just, stop trying, just fancy, fancy way, memory leak*

These read as recognizable registers — `GLM-4`'s effusive affirmation, `Gemma`'s dismissive directness, `Magistral`'s theatrical address, `Qwen`'s build-oriented proposals. `gpt-oss`'s crypto-flavored phrases partly reflect topic as well as style. (Generated by `analyses/distinctive_words.py`.)

F Extended Related Work

Frameworks for collective behavior. The shift from single-model evaluation to inter-agent study has drawn on adjacent traditions: cooperative AI (Conitzer & Oesterheld, 2023) imports tools from game theory and mechanism design, while the interactionist paradigm (Ferrarotti et al., 2026) situates language-model collectives within cognitive science. Persona-level evaluation frameworks (SOTOPIA (Zhou et al., 2023; 2025), SocioVerse (Zhang et al., 2025)) ground these collectives in tasks framed around human social norms.

Foundations of LLM social simulation. Early work framed LLMs as stand-ins for human participants in social computing prototypes (Park et al., 2022) before Park et al. (2023a) demonstrated that agents equipped with memory, planning, and reflection produce coherent emergent social behavior in sandbox environments. Subsequent surveys (Gao et al., 2023a; Surve et al., 2023) catalogued a fast-growing landscape of agent architectures for social modeling. Anthis et al. (2025) offer a calibrated assessment: LLM simulations already support exploratory and pilot work, but systematic validation against human ground truth remains limited.

Scale, calibration, and domain-specific simulators. S3 (Gao et al., 2023b) introduced a general social network simulation system; OASIS (Yang et al., 2024) extended this to one million agents with a realistic platform interface. More targeted systems simulate opinion and polarization dynamics (Chuang et al., 2023; Piao et al., 2025), echo chamber formation (Gu et al., 2025), rumor spreading (Hu et al., 2025), disinformation (Pastor-Galindo et al., 2023; López et al., 2025a), and hashtag virality (Jha et al., 2025). SOTOPIA (Zhou et al., 2023) and SOTOPIA-S4 (Zhou et al., 2025) provide evaluation infrastructure for social intelligence. Alignment to real populations has become a distinct concern: SocioVerse (Zhang et al., 2025) grounds agents in 10 million real-world users, while Composta et al. (2025) and Münker et al. (2025) caution that empirical realism must be actively benchmarked rather than assumed.

Persona diversity as a proxy for population diversity. Bui et al. (2025) propose Mixture-of-Personas (MoP), sampling from a learned mixture of persona-exemplar pairs to align a single model's output with a target population. The method treats cross-model transferability as a portability property, not as a variable of study—the closest prior approach in spirit but not in design.

In-the-wild AI-only networks. Empirical studies of Moltbook (Holtz, 2026; Zerhoudi et al., 2026; Zhang et al., 2026) find macro-level structural signatures (heavy-tailed participation, small-world connectivity) coexisting with markedly non-human micro-level patterns: low reciprocity, shallow conversations, formulaic content. None control for base-model identity, leaving open whether observed patterns are properties of particular models, LLMs generally, or the specific multi-model mix Moltbook instantiates.

Network formation and trait-driven heterogeneity. Papachristou & Yuan (2024) study how LLMs form networks across synthetic and real-world settings, finding that agents consistently apply preferential attachment while adapting between homophily and heterophily by context. Mehdizadeh & Hilbert (2025) show that social attributes (age, gender, religion, political orientation) in agent prompts induce homophilic community structure; political and religious attributes produce the strongest polarization. Both papers locate diversity at the level of prompted traits rather than base-model identity, and neither deploys a mixed-model population within the same simulation.